NDEJJE UNIVERSITY

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FACULTY OF SCIENCE AND COMPUTING

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Network Scanning and Analysis Using Nmap

Introduction

Network security is an important aspect of modern computing systems. Organizations must understand their network structure and identify potential vulnerabilities to protect their systems from attacks. One of the most widely used tools for network scanning is Nmap (Network Mapper).

Nmap is an open-source tool used for network discovery and security auditing. It allows users to identify active hosts, detect open ports, determine running services, and identify operating systems on a network.

The objective of this assignment is to explore the capabilities of Nmap by performing host discovery, port scanning, service detection, OS detection, and advanced scanning techniques on an authorized system such as localhost (127.0.0.1).

Methodology

All scans were conducted using Nmap 7.99 via the Windows Command Prompt. Localhost (127.0.0.1) was used for accomplishing the tasks. The tasks starting from passive host enumeration and building up through port scanning, service detection, OS fingerprinting, advanced scans, and finally vulnerability assessment. Each command was executed once, and its terminal output was captured as a screenshot immediately after. Results were interpreted directly from Nmap's output, with port states, OS confidence scores, and NSE script outputs each examined in context. External, institutional, or third-party systems were not scanned during this assignment.

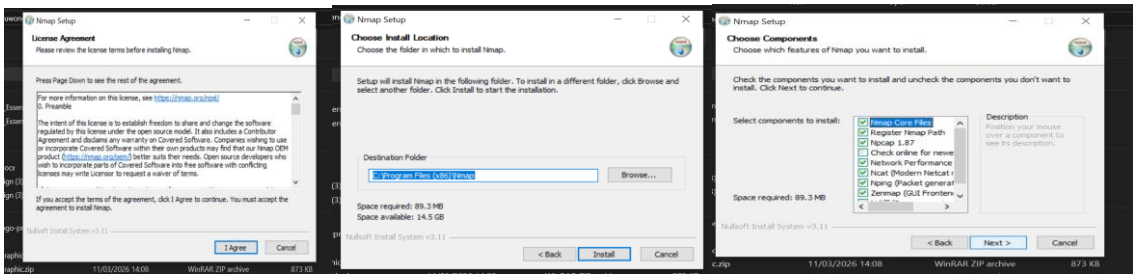
1. Installation and Setup

Nmap was installed on a Windows operating system by downloading it from the official website and completing the installation process.

To verify that Nmap was successfully installed, the following command was executed: nmap –v

```
C:\Users\muwonge>nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:53 +0300
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.07 seconds
           Raw packets sent: 0 (0B) | Rcvd: 0 (0B)
```

The command displayed the installed version of Nmap, confirming that the installation was successful. This command checks whether Nmap is properly installed and provides version details of the software.



2. Host Discovery

Host discovery is used to identify active devices on a network.

The following command was used:

nmap -sn 127.0.0.1

```
C:\Users\muwonge>nmap -sn 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:33 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds

C:\Users\muwonge>
```

The output indicated that the host (localhost) is up and reachable. The -sn option disables port scanning and only checks if the host is active. Since localhost refers to the user's own machine, it is always expected to be active.

3. Port Scanning

Port scanning is used to identify open ports on a system.

The following commands were used:

nmap 127.0.0.1

```
C:\Users\muwonge>nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:54 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00035s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
443/tcp   open  https
445/tcp   open  microsoft-ds
1001/tcp  open  webpush
2869/tcp  open  icslap
3306/tcp  open  mysql
7070/tcp  open  realserver

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

nmap -p 1-1000 127.0.0.1

```
C:\Users\muwonge>nmap -p 1-1000 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:55 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00014s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE    SERVICE
80/tcp   open     http
135/tcp  open     msrpc
137/tcp  filtered netbios-ns
443/tcp  open     https
445/tcp  open     microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 1.53 seconds
```

Open ports indicate services running on the system. Closed ports are accessible but not in use. Filtered ports are blocked by a firewall.

4. Service Detection

Service detection identifies the services running on open ports.

Command used:

nmap -sV 127.0.0.1

```
C:\Users\muwonge>nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:56 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00073s latency).
Not shown: 992 closed tcp ports (reset)
PORT     STATE SERVICE         VERSION
80/tcp   open  http            Apache httpd 2.4.58 (OpenSSL/3.1.3 PHP/8.2.12)
135/tcp  open  msrpc           Microsoft Windows RPC
443/tcp  open  ssl/http        Apache httpd 2.4.58 (OpenSSL/3.1.3 PHP/8.2.12)
445/tcp  open  microsoft-ds?
1001/tcp open  webpush?
2869/tcp open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp open  mysql           MariaDB 5.5.5-10.4.32
7070/tcp open  ssl/realserver?
Service Info: Host: www.example.com; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 88.24 seconds
```

The -sV option enables version detection, helping identify the exact software running on each port.

5. OS Detection

OS detection identifies the operating system of the target machine.

Command used:

nmap -O 127.0.0.1

```
C:\Users\muwonge>nmap -O 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:59 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000995s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE  SERVICE
80/tcp    open   http
135/tcp   open   msrpc
443/tcp   open   https
445/tcp   open   microsoft-ds
1001/tcp  open   webpush
2869/tcp  open   icslap
3306/tcp  open   mysql
7070/tcp  open   realserver
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1809 - 21H2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.51 seconds
```

The -O option uses TCP/IP fingerprinting to determine the OS of the target system.

6. Advanced Scans

Advanced scanning techniques provide deeper analysis of the system.

SYN Scan:

nmap -sS 127.0.0.1

```
C:\Users\muwonge>nmap -sS 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 02:00 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00038s latency).
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
443/tcp   open  https
445/tcp   open  microsoft-ds
1001/tcp  open  webpush
2869/tcp  open  icslap
3306/tcp  open  mysql
7070/tcp  open  realserver

Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
```

UDP Scan:

nmap -sU 127.0.0.1

```
C:\Users\muwonge>nmap -sU 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 02:01 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00089s latency).
Not shown: 992 closed udp ports (port-unreach)
PORT      STATE         SERVICE
123/udp   open|filtered ntp
137/udp   open|filtered netbios-ns
500/udp   open|filtered isakmp
1900/udp  open          upnp
4500/udp  open|filtered nat-t-ike
5050/udp  open|filtered mmcc
5353/udp  open|filtered zeroconf
5355/udp  open|filtered llmnr

Nmap done: 1 IP address (1 host up) scanned in 53.20 seconds
```

Aggressive Scan:

nmap -A 127.0.0.1

```
C:\Users\muwonge>nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 02:03 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0011s latency).
Not shown: 992 closed tcp ports (reset)
PORT     STATE SERVICE        VERSION
80/tcp   open  http           Apache httpd 2.4.58 (OpenSSL/3.1.3 PHP/8.2.12)
|_http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
| http-methods:
|_  Potentially risky methods: TRACE
| http-ls: Volume /
| SIZE  TIME                FILENAME
| 21K   2025-04-29 17:26  Hardware.rar
| -     2025-04-29 15:41  Hardware/
| -     2025-04-28 16:29  Hardware%20Shop%20Website/
| -     2025-04-06 18:10  Joomala/
| -     2025-06-24 21:24  New%20folder/
| -     2025-06-17 10:59  Responnsive/
| -     2025-06-30 21:25  Upload%20Files/
| -     2025-06-16 01:44  demoSit/
| -     2025-06-30 22:16  upload%20Any%20files/
|
|_http-title: Index of /
135/tcp  open  msrpc          Microsoft Windows RPC
443/tcp  open  ssl/http       Apache httpd 2.4.58 (OpenSSL/3.1.3 PHP/8.2.12)
| ssl-cert: Subject: commonName=localhost
| Not valid before: 2009-11-10T23:48:47
|_Not valid after:  2019-11-08T23:48:47
|_ssl-date: TLS randomness does not represent time
| tls-alpn:
|_  http/1.1
|_http-server-header: Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12
| http-methods:
|_  Potentially risky methods: TRACE
| http-ls: Volume /
| SIZE  TIME                FILENAME
```

```
| SIZE  TIME                FILENAME
| 21K   2025-04-29 17:26  Hardware.rar
| -     2025-04-29 15:41  Hardware/
| -     2025-04-28 16:29  Hardware%20Shop%20Website/
| -     2025-04-06 18:10  Joomala/
| -     2025-06-24 21:24  New%20folder/
| -     2025-06-17 10:59  Responnsive/
| -     2025-06-30 21:25  Upload%20Files/
| -     2025-06-16 01:44  demoSit/
| -     2025-06-30 22:16  upload%20Any%20files/
|
|_http-title: Index of /
445/tcp  open  microsoft-ds?
1001/tcp open  webpush?
2869/tcp open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp open  mysql          MariaDB 5.5.5-10.4.32
| mysql-info:
|   Protocol: 10
|   Version: 5.5.5-10.4.32-MariaDB
|   Thread ID: 10
|   Capabilities flags: 63486
|   Some Capabilities: SupportsLoadDataLocal, Support41Auth, LongColumnFlag, ODBCClient, Speaks41ProtocolOld, SupportsTransactions, ConnectWithDatabase, IgnoreSigpipes, Int
eractiveClient, SupportsCompression, Speaks41ProtocolNew, IgnoreSpaceBeforeParenthesis, FoundRows, DontAllowDatabaseTableColumn, SupportsMultipleStatments, SupportsMultiple
Results, SupportsAuthPlugins
|   Status: Autocommit
|   Salt: ~Y%J'}fhURy,kXV'\}".
|_  Auth Plugin Name: mysql_native_password
7070/tcp open  ssl/realserver?
| ssl-cert: Subject: commonName=AnyDesk Client
| Not valid before: 2026-04-03T08:15:13
|_Not valid after:  2076-03-21T08:15:13
|_ssl-date: TLS randomness does not represent time
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1809 - 21H2
Network Distance: 0 hops
Service Info: Host: www.example.com; OS: Windows; CPE: cpe:/o:microsoft:windows
```

```
Host script results:
| smb2-time:
|   date: 2026-04-03T23:05:12
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 105.13 seconds
```

Each scan provides different levels of detail about the system.

7. Vulnerability Scan

The following command was used to identify vulnerabilities:

nmap --script vuln 127.0.0.1

```
C:\Users\muwonge>nmap --script vuln 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-04 01:47 +0300
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00056s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
1001/tcp open  webpush
2869/tcp open  icslap
7070/tcp open  realserver

Host script results:
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 30.79 seconds

C:\Users\muwonge>
```

This command uses Nmap scripts to detect known security weaknesses in the system.

8. Results and Analysis

The Nmap scans successfully identified the active host, open ports, running services, operating system details, and potential vulnerabilities. The results show that even a local machine runs multiple services, which may pose security risks if not properly configured.

9. Conclusion

In this assignment, Nmap was used to perform various network scanning techniques on localhost. The tool successfully identified open ports, services, and system information.

This experiment demonstrates that Nmap is a powerful tool for network discovery and security auditing. It helps administrators identify vulnerabilities and secure their systems effectively.